BINAURAL VOCAL-BAND DYNAMIC SNR, SUBBAND ENVELOPE CORRELATION, AND LEXICAL PREDICTABILITY FOR LYRICS INTELLIGIBILITY PREDICTION IN CLIP2

ICASSP 2027 Signal Processing Grand Challenge — 2nd Cadenza Lyrics Intelligibility Prediction Challenge (CLIP2)
***System Classification:** Multimodal Hybrid Intrusive & Non-Intrusive Model (Audio1 + Audio2 + Severity + Lyrics)*

1. ABSTRACT

Predicting sung lyrics intelligibility for listeners with varying degrees of sensorineural hearing loss (*No Loss*, *Mild*, and *Moderate*) requires jointly modeling peripheral auditory degradation, instrumental accompaniment masking in the vocal formant region, and top-down linguistic predictability. We present a lightweight, interpretable, physics-informed psychoacoustic and lexical regression framework for the 2nd Cadenza Lyrics Intelligibility Prediction Challenge (CLIP2). Operating across all four officially permitted inference modalities—**Audio1** (hearing-loss-simulated binaural stereo FLAC), **Audio2** (unprocessed reference binaural stereo FLAC), **Severity** (hearing-loss tier), and ground-truth **Lyrics** prompts—our model extracts 14 binaural and lexical descriptors anchored by: (1) Vocal-Band Dynamic SNR Crest (450–3,600 Hz 90th-to-20th percentile short-time energy range), (2) High-Frequency Accompaniment Spectral Centroid masking, (3) Better-Ear Subband Short-Time Envelope Correlation (STOI-Vocal) between aligned Audio1 and Audio2, and (4) Lexical Predictability (average word length in characters, closed-class function word ratio, and log utterance length). On Leave-One-Out Cross-Validation (LOO-CV) over the official 44.1 kHz stereo FLAC training benchmark, our system reduces Root-Mean-Square Error (RMSE) from **41.25%** (Constant Mean Baseline) to **24.26%** (a **41.2% relative error reduction**) while requiring zero external pretrained weights and < 0.35 seconds CPU runtime per musical excerpt.

2. MULTIMODAL SYSTEM ARCHITECTURE & PSYCHOACOUSTIC FEATURE ENGINEERING

2.1. Binaural Group-Delay Alignment & Subband Envelope Correlation (Audio1 vs. Audio2)

Hearing-loss simulation pipelines introduce frequency-dependent group delay between the impaired binaural signal $x^{(1)}_{L,R}(t)$ (Audio1) and the unprocessed reference signal $x^{(2)}_{L,R}(t)$ (Audio2). For each stereo channel c in {L, R}, we downsample from 44.1 kHz to 16 kHz via polyphase antialiasing filtering and align $x^{(1)}_c$ and $x^{(2)}_c$ using FFT cross-correlation over the central segment (±640 samples lag search). We then decompose both aligned signals across five 3rd-order Butterworth bandpass subbands spanning vocal fundamental, formant, and fricative/stop consonant frequencies:

```
B = { [200, 450], [450, 900], [900, 1800], [1800, 3600], [3600, 6800] } Hz
```

Using 25 ms short-time root-mean-square energy frames with a 10 ms hop (100 Hz frame rate), we compute the log-envelope Pearson correlation $r_{c,b}$ in each subband b and extract both the better-ear maximum $\max_c$ (1/5 Sum_b $r_{c,b}$) and the binaural mean, together with high-frequency consonant energy retention (1.8–6.8 kHz vs. 200–900 Hz energy ratio in dB between Audio1 and Audio2).

2.2. Vocal-Band Dynamic SNR Crest & Accompaniment Spectral Masking (Audio1)

Unlike speech in stationary noise, sung vocals in popular and rock music suffer from severe energetic and modulation masking when dense instrumental accompaniment fills the temporal dips between syllables. On the core vocal formant passband (450–3,600 Hz), we extract the 100 Hz short-time energy envelope $E_v(m)$ and define the **Vocal-Band Dynamic SNR Crest**:

```
Crest_vocal = P_90( 10 log10 E_v^2 ) - P_20( 10 log10 E_v^2 ) [dB]
```

When lead vocals emerge clearly above the instrumental accompaniment floor, Crest_vocal is large (20–34 dB), exhibiting the strongest positive correlation with human word recognition rate on the CLIP2 benchmark (r = +0.493). Conversely, high better-ear spectral centroid $f_{centroid}$ (> 1.8 kHz, driven by loud cymbals, hi-hats, or distorted guitars) masks high-frequency consonant cues and negatively correlates with intelligibility (r = -0.308). We additionally compute the stereo Mid-to-Side vocal ratio $10 \log_{10}(E_{Mid} / E_{Side})$ in 300–3,400 Hz and the 2–8 Hz syllabic envelope modulation power ratio.

2.3. Top-Down Lexical Predictability & Phonemic Confusability (Lyrics & Severity)

Human listeners actively exploit lexical familiarity and syntactic context to reconstruct partially masked phonemes. Because CLIP2 rules explicitly allow using the ground-truth *Lyrics* prompt and *Severity* metadata at validation and evaluation time, we extract four linguistic descriptors from the target utterance $W = (w_1, ..., w_M)$:

- **Average Word Length (char_per_word):** Mean characters per word, strongly negatively correlated with human intelligibility (r = -0.483) because longer, lower-frequency content words are harder to decode under acoustic masking.
- **Closed-Class Function Word Ratio (function_word_ratio):** Fraction of high-frequency English function words (*the, a, to, in, of, my, you, me, i, it, is, and, ...*), positively correlated with intelligibility (r = +0.264).

- **Contextual Utterance Length (log_n_words):** log(1 + *M*), capturing sentence-level semantic constraint (*r* = +0.239), paired with stop/fricative consonant density (*s*, *f*, *t*, *k*, *p*, *v*, *z*).

3.EMPIRICAL FEATURE CORRELATIONS & REGULARIZED HYBRID CALIBRATION

Table 1 summarizes the 14 extracted binaural psychoacoustic and lexical features and their empirical Pearson correlation with ground-truth listener word recognition scores (0–100%) on the official 44.1 kHz stereo FLAC benchmark.

Idx	Feature Name	Modality Source	Range / Units	Corr (r) vs. Correctness
0–2	hl_no_loss / hl_mild / hl_moderate	Severity Metadata	{0, 1} One-Hot	-0.416 / +0.486 / +0.092
3–4	better_ear_stoi_corr / binaural_stoi_mean	Audio1 + Audio2	[0.91, 1.00]	-0.351 / -0.360
5	hf_consonant_retention_db (1.8–6.8 kHz)	Audio1 + Audio2	[-16.0, 0.0] dB	-0.189
6–7	vocal_mid_to_side_db / syllabic_4hz_mod	Audio1 (Stereo)	[-1.2, 10.4] dB	-0.212 / -0.246
8	dynamic_snr_crest_db (450–3,600 Hz P90-P20)	Audio1 (Vocal Band)	[4.2, 34.4] dB	+0.493 (Strongest +)
9	spectral_centroid_khz (Accompaniment Mask)	Audio1 (Better Ear)	[0.68, 2.80] kHz	-0.308 (Masking -)
10	log_n_words (Utterance Context Length)	Lyrics Prompt	[1.95, 2.30]	+0.239
11	char_per_word (Lexical Complexity)	Lyrics Prompt	[2.78, 5.00] chars	-0.483 (Strongest -)
12–13	stop_fricative_ratio / function_word_ratio	Lyrics Prompt	[0.00, 0.56]	+0.001 / +0.264

3.1. Calibrated Psychoacoustic Prior + L2 Ridge Residual Regression

To prevent overfitting across unseen musical genres and hearing-loss configurations while adapting smoothly from small demo splits to the full 6,000-signal training corpus, we combine a bounded domain-knowledge psychoacoustic prior $y_prior(\mathbf{x})$ in [0, 100] with an L2-regularized Ridge residual estimator fitted on standardized features **x_norm** = (**x** - **mu**) / **sigma**:

```
y_hat(x) = clip_[0, 100]( y_prior(x) + gamma * x_norm^T w_Ridge ), w_Ridge = (X_norm^T X_norm +
                                    alpha I)^(-1) X_norm^T (y - y_prior)
```

where adaptive regularization sets (alpha = 45.0, gamma = 0.25) when $N < 50$ and (alpha = 15.0, gamma = 0.60) on full training splits. All predictions are scaled to [0, 100] percentage points as required by both EvalAI (#2703) and the ICASSP 2027 Grand Challenge evaluation protocol.

4. VALIDATION BENCHMARK RESULTS & ABLATION STUDY

Table 2 reports Leave-One-Out Cross-Validation (LOO-CV) Root-Mean-Square Error (RMSE, in percentage points) across progressive modality ablations on the official stereo 44.1 kHz FLAC training benchmark, alongside sample validation predictions generated for the EvalAI (#2703) and ICASSP 2027 submission files.

System Configuration	Permitted Modalities Used	External Weights	LOO-CV RMSE (%)	Relative Gain vs. Ba
Constant Mean Baseline	None	No	41.25%	—
Severity + Lexical Only	Severity + Lyrics	No	33.40%	-19.0%
Binaural Psychoacoustic Only	Audio1 + Audio2 + Severity	No	31.84%	-22.8%
Proposed Binaural STOI-Vocal + Lexical Solver	**Audio1 + Audio2 + Severity + Lyrics**	**No**	**24.26%**	**-41.2% (-16.99%p)**

5.COMPLEXITY, CAUSALITY & COMPLIANCE STATEMENT

Our complete inference pipeline (`cadenza_clip2_solver.py`) is 100% deterministic, uses only `numpy`, `scipy`, and `soundfile` (zero external neural checkpoints or third-party lyrics datasets), processes each 44.1 kHz stereo FLAC pair in ~0.28 seconds on a single CPU core, and automatically outputs both the headerless EvalAI CSV (`evalai_valid_submission.csv`) and the headered ICASSP 2027 CSVs (`ICASSP2027_valid_T001.csv` and `ICASSP2027_eval_T001.csv`).